

Carbon-Aware Peer-to-Peer Energy Trading in an Unbalanced Distribution Network via a Nash Equilibrium Discovery Deep Reinforcement Learning Approach

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Abstract—Peer-to-peer (P2P) energy trading offers an innovative platform for prosumers to engage in bilateral electricity and carbon trades, enabling promoting the integration of distributed energy resources and supporting low-carbon operations in distribution networks. However, existing studies often rely on single-phase approximations, leading to inaccuracies in emission accounting and potential economic inefficiencies loss. To address these challenges, this paper proposes a carbon-aware P2P energy trading mechanism for unbalanced distribution networks. A novel joint electricity and carbon trading market framework is developed, integrating carbon permits trading with a continuous double auction mechanism. The trading process is formulated as a cooperative Markov game to capture the stochastic nature of market dynamics and to enable intelligent bidding decisions. A Nash equilibrium (NE) discovery multi-agent deep reinforcement learning framework is employed, utilizing a trust region policy optimization-based algorithm to enhance trading reliability and policy interpretability. Simulations on a modified IEEE-123 unbalanced distribution network demonstrate the proposed effectiveness in reducing operational costs and carbon emissions, providing rigorous NE validation and showcasing its practical applicability in complex energy markets.

Index Terms—Carbon emission flow, distribution network, Nash equilibrium, multi-agent deep reinforcement learning, peer-to-peer energy trading, unbalanced distribution network.

I. INTRODUCTION

TO FACILITATE the low-carbon transition of the power system, there has been a significant increase in distributed energy resources (DERs), renewable energy sources (RES),

energy storage systems (ESSs), and flexible loads (FLs) within distribution networks in recent years. This growth has led to the rise of a new type of entity called prosumers and a new trading paradigm, peer-to-peer (P2P) energy trading, which offers substantial benefits to the participating prosumers and the distribution system operator (DSO).

P2P energy trading can provide benefits through multi-bilateral negotiation methods with auction models [1], game theory [2], or optimization models [3]. While the network constraints were overlooked [4], [5], the feasibility of a market trading outcome must be confirmed by the DSO, in order to safeguard the secure operation of the distribution network. In [6], the authors consider the physical network for P2P energy trading under a transactive energy framework. Reference [7] proposes P2P energy trading for microgrids at a lower level, the network operating conditions are supervised by a DSO at an upper level. Furthermore, the distribution network operating constraints can be guaranteed by optimization methods in [8], [9], [10], [11], where the consensus needs to be achieved among prosumers and the DSO takes the duty for operating safety.

However, the majority of network-constrained P2P trading studies assume a single-phase equivalent distribution network model, whereas the distribution networks are three-phase unbalanced in nature and the integration of prosumers with DER can introduce various operational problems such as voltage violation and phase unbalances in the network. Consequently, it is of pivotal importance to integrate network constraints in the P2P energy trading and market clearing models with an accurate unbalance distribution network model.

On the other hand, environment-friendly operation is also critical for energy market clearing. According to the report of CSIRO in 2022 [12], although greenhouse emissions have declined due to renewable energy resources, electricity generation still accounts for about three-fourths of total emissions. Therefore, the DSO and P2P platforms should simultaneously consider economic benefits and ecological impacts. Consequently, it is urgent to coordinate the carbon emission auction market and the P2P energy market to effectively

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promote emission reduction by providing financial incentives. Essentially, the carbon emission auction market can serve as a trading platform for carbon emission permits distributed by the market operator to all prosumers. If a prosumer's emissions surpass their allocated permits, they must purchase additional permits in the auction market to avoid severe penalties.

To this end, carbon emission accounting [13] aims to evaluate ecological performance, and it can be integrated into an energy trading framework for carbon-aware trading. Recently, [14] proposes a P2P carbon allowance trading scheme in a distribution network, while [15] resorts to carbon emission caps as a constraint for prosumers. Further, [16] develops joint electricity and carbon trading (JECT) for multiple microgrids, and [17] proposes a JECT market. Although these studies have considered carbon-related markets across various entities, two outstanding challenges remain. Firstly, for the prosumers with changeable roles (i.e., whether they act as producers or consumers), accurately quantifying their carbon emissions from a system-level perspective is essential, given inaccurate quantification may lead to an unfair and less economic trading outcome, discouraging prosumers' effort for carbon reduction. Secondly, it is time to address integrating the carbon auction market into a P2P electricity market without sacrificing flexibility and scalability.

For the former, carbon emission flow (CEF) [18] can efficiently trace the carbon emission from the generation side to the demand side with the power system operation. It provides fairness in carbon emission accounting for the prosumers. However, the recursive CEF algorithm brings nonlinearity and leads to intractable solutions for the energy market clearing process.

For the latter, the current electricity P2P trading market can be categorized into coordinated market, decentralized market and community market. The first two markets suffer from either computation burdens or optimality concerns, as summarized by [19]. The community market, e.g., the continuous double auction (CDA) market studied in [19] as a potential solution, takes advantage of the first two types and can provide high computation efficiency and optimality. However, the JECT is highly stochastic in the CDA market, and the market dynamics cannot be well captured with limited information, necessitating the participating prosumers to equip suitable decision-making tools that can promptly react to the fast-evolving market dynamics.

In light of the above challenges of nonlinear and stochastic market dynamics adaptability while pursuing decision-making efficiency, deep reinforcement learning (DRL) constitutes a suitable methodological framework that can support online decision-making for energy trading price and quantity by interacting with the market clearing environment. Since P2P energy trading is a multi-agent problem, the naïve approach is to extend a DRL approach to a corresponding multi-agent DRL (MADRL) algorithm. MADRL has been intensively studied in P2P trading problems [5], [20], [21], [22], [23], [24], [25]. However, Nash equilibrium (NE) criteria, the critical performance metric for the MADRL outcome that governs the efficiency of the JECT market, has not been examined in the majority of MADRL-based P2P trading studies. Intuitively,

the prosumers cannot determine the 'optimal' bidding decision at the start of P2P negotiation, since the strategy of one prosumer is impacted by the unknown variables and strategies of others. Therefore, prosumers adapt and modify their strategies and eventually converge to a steady state where no prosumer can obtain more profits by changing their bids if others' bids remain fixed. Recently, the literature either estimates [17] or approximates [26] NE conditions with a MADRL algorithm, while the discovery of NE solution discovery is still an open question that needs further investigation.

This work aims to address the aforementioned research challenges in terms of network model accuracy, CEF recursive algorithm, JECT uncertainty, and the proof of NE. The main contributions of this work are summarized as follows:

- 1) A bi-level carbon-aware P2P energy trading model is formulated with a CDA market. The upper level represents the distribution network operation, which ensures the network is responsible for ensuring network security and clearing excessive carbon emission prices. The lower level consists of P2P agents that submit joint electricity and carbon trading bids to the CDA market. The whole model is reformulated as a Markov Game to be compatible with an application of online data-driven optimization method.
- 2) Unlike recent P2P energy trading studies that overlook network models with single-phase equivalent assumptions, this paper proposes a carbon-aware optimal power flow model with an effective three-phase unbalanced network model. This model can accurately account for carbon emissions for prosumers on a '*nodal-phase*' level. Additionally, it provides service pricing through distributed locational marginal prices for carbon, enabling prosumers to efficiently reduce their carbon emissions.
- 3) Unlike existing MADRL methods for P2P energy trading, lacking the NE analysis. Inspired by the finding in [27], this paper applies an NE-discovery MADRL method based on a trust region policy optimization algorithm. The method employs a sequential policy update within the training framework, ensuring policy interpretability and the convergence of P2P energy trading with proven NE discovery.

II. JOINT ELECTRICITY AND CARBON TRADING FRAMEWORK

This paper proposes a JECT framework including physical, market and coordination layers which are introduced below in Fig. 1.

- 1) In the physical layer, each prosumer equipped with DERs is connected through a specific phase in an unbalanced distribution network. The distribution network comprises a cluster of buses and lines under three-phase configurations.
- 2) In the P2P trading market layer, each prosumer submits bids for both electricity and carbon quantities and prices in a CDA market. Any unmatched quantities are cleared in a peer-to-grid (P2G) market with the DSO as a similar

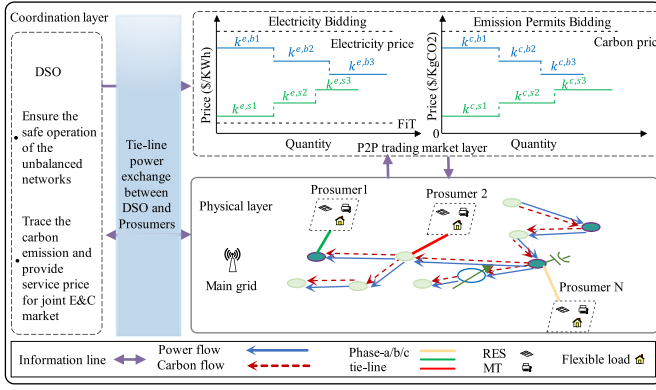


Fig. 1. The trading framework.

market settlement [28]. Based on bids, the auctioneer determines the internal P2P energy trading quantity and prices within the CDA market, while the DSO clears the P2G prices.

- 3) In the coordination layer, the interaction among the DSO and prosumers is formulated as a bi-level optimization model, seeking an agreed optimal trading result. The DSO's upper-level operational model aims to minimize costs considering carbon emissions and using a carbon aware optimal power flow (CAOPF) model, which ensures network operating constraints and provides electricity/carbon prices to the lower level. At the lower level, prosumers aim to minimize their carbon aware operational costs through local energy management.

III. CARBON-AWARE P2P TRADING MODEL

A. Distribution Network Operation Model

The DSO provides two services for carbon-aware P2P trading. Firstly, the safe operation of the unbalanced network should be ensured when carbon-aware P2P trading is integrated into the coordination. Secondly, it can provide carbon-aware service prices and emission accounting for P2P agents. The DSO model consists of network power flow and carbon emission constraints as a carbon-aware market-clearing model. Considering an unbalanced radial distribution network with $N + 1$ buses, the root bus is indexed as 0, and $\mathcal{N} = \{1, 2, \dots, N\}$ for the remaining buses in the system. The branch set is denoted by $\mathcal{E} = \{l = (i, j) \mid i = \text{sup}(j)\}$, $\text{sup}(j)$ represents the superior node for node j . The downstream bus of node j is indexed, where $k = \text{down}(j)$. The phase set for the unbalanced network can be written as $\Phi = \{a, b, c\}$, and the phase is indexed as φ . For each bus j and branch $ij \in \mathcal{E}$, the following vectors can be defined voltage magnitude $V_{j,t} = [V_{j,t}^a, V_{j,t}^b, V_{j,t}^c]^T$, active line flow $P_{ij,t} = [P_{ij,t}^a, P_{ij,t}^b, P_{ij,t}^c]^T$, reactive power line flow $Q_{ij,t} = [Q_{ij,t}^a, Q_{ij,t}^b, Q_{ij,t}^c]^T$ and the squared current $L_{ij,t} = [L_{ij,t}^a, L_{ij,t}^b, L_{ij,t}^c]^T$. As similar to the previous formatting the variable in unbalanced network, the resistance is written as $r_{ij} = [r_{ij}^a, r_{ij}^b, r_{ij}^c]^T$, the reactance is written as $x_{ij} = [x_{ij}^a, x_{ij}^b, x_{ij}^c]^T$. We also introduce the power exchange at

the tie-line as the variable $P_{j,t}^{\text{tie}} = [P_{j,t}^{\text{tie},a}, P_{j,t}^{\text{tie},b}, P_{j,t}^{\text{tie},c}]^T$. The prosumer set is denoted as Ω_{prosumer} , the element in this set is indexed as b . The node j or phase φ connected with prosumer b is written as $j(b), \varphi(b)$. In our notation, DERs output with uppercase power symbols denotes DSO owned device (e.g., $P_{j,t}^{\text{Res},\varphi}$), while DERs output lowercase power symbols represent DERs owned by prosumers (e.g., $p_{b,t}^{\text{Res}}$).

1) *Unbalanced Distribution Network Model*: The unbalanced power flow constraints are modeled with extension from the widely used branch flow model [29]. According to Kirchhoff's current law and the impedance matrix $z_{ij} = r_{ij} + jx_{ij}$, where $z_{ij} \in \mathbb{R}^{3 \times 3}$, the active and reactive power can be written as below:

$$P_{ij,t} - r_{ij}L_{ij,t} = \sum_{k \in \text{down}(j)} P_{jk,t} + P_{j,t}^{\text{tie}}, \forall j, \forall t \quad (1)$$

$$Q_{ij,t} - x_{ij}L_{ij,t} = \sum_{k \in \text{down}(j)} Q_{jk,t} + Q_{j,t}^L, \forall j, \forall t \quad (2)$$

$$P_{j,t}^{\text{tie},\varphi} = p_{j(b),t}^{\text{net},\varphi(b)} + P_{j,t}^{L,\varphi} - P_{j,t}^{\text{Res},\varphi} - P_{j,t}^{\text{MT},\varphi}, \quad \forall j, \forall t, \forall \varphi \quad (3)$$

Equation (3) denotes the active power exchange at the phase φ , node j at tie-line $P_{j,t}^{\text{tie},\varphi}$. The netload $p_{j(b),t}^{\text{net},\varphi(b)}$ for prosumer b will be calculated in the following chapter. The parameters $P_{j,t}^{L,\varphi}$, $P_{j,t}^{\text{Res},\varphi}$ and $P_{j,t}^{\text{MT},\varphi}$ are the load, RES and micro turbine (MT) in the distribution network besides the prosumer.

Then the voltage drop constraints can be modeled for $\forall ij$ as:

$$\begin{aligned} V_{j,t} &= V_{i,t} - z_{ij}L_{ij,t} \\ &= V_{i,t} - z_{ij}[(P_{ij,t} - jQ_{ij,t}) \oslash V_{i,t}^*] \end{aligned} \quad (4)$$

In equation (4), the $\oslash$ indicates the **element-wise division** for vectors computation. Due to phase coupling, which arises from the cross-products of three-phase terms in equation (4), the formulation exhibits nonlinearity. Therefore, we assume that bus voltages are nearly balanced as the following approximation:

$$V_j^a/V_j^b \approx V_j^b/V_j^c \approx V_j^c/V_j^a \approx e^{j2\pi/3} \quad (5)$$

Then, the $V_{j,t}$ can be reformed as:

$$\begin{aligned} V_{j,t} &= [V_j^a \ V_j^b \ V_j^c]^T \\ &= [V_j^a \ |V_j^b|e^{-j2\pi/3} \ |V_j^c|e^{j2\pi/3}]^T \end{aligned} \quad (6)$$

With the multiplication with their conjugates [30], the voltage drop constraints are reformulated with the squared voltage $(U_j = [U_j^a \ U_j^b \ U_j^c]^T)$ as follows:

$$U_{j,t} = U_{i,t} - 2(\tilde{r}_{ij}P_{ij,t} + \tilde{x}_{ij}Q_{ij,t}) + \tilde{z}_{ij}L_{ij,t}, \forall ij, t \quad (7)$$

The phase coupling parameters $\tilde{r}_{ij}$, $\tilde{x}_{ij}$ and $\tilde{z}_{ij}$ in (7) are calculated with the following equations (8)–(11), the $\oslash$ denotes element-wise multiplication, $\text{Re}\{\}$ indicates the real part and $\text{Im}\{\}$ denotes the imaginary part:

$$\tilde{r}_{ij} = \text{Re}\{\alpha \oslash r_{ij}\} - \text{Im}\{\alpha \oslash x_{ij}\} \quad (8)$$

$$\tilde{x}_{ij} = \text{Re}\{\alpha \oslash x_{ij}\} + \text{Im}\{\alpha \oslash r_{ij}\} \quad (9)$$

$$\tilde{z}_{ij} = |z_{ij}|^{\odot 2} \quad (10)$$

$$\alpha = \begin{bmatrix} 1 & e^{-j2\pi/3} & e^{j2\pi/3} \\ e^{j2\pi/3} & 1 & e^{-j2\pi/3} \\ e^{-j2\pi/3} & e^{j2\pi/3} & 1 \end{bmatrix} \quad (11)$$

Then, the voltage magnitude and current constraints can be formulated as:

$$(\underline{V})^2 \leq U_{j,t} \leq (\overline{V})^2, \forall j, t \quad (12)$$

$$[(P_{ij,t})^{\odot 2} + (Q_{ij,t})^{\odot 2}] \leq L_{ij,t} \odot U_j, \forall ij, \forall j \quad (13)$$

where $[\underline{V}, \overline{V}]$ is the lower and upper boundary in (12). Equation (13) indicates the second-order conic programming constraints; the general inequality form for this formulation can be found in [31]. In summary, the unbalanced network power flow constraints include (1)–(3), (7), (12) and (13). The abovementioned model in (1)–(13) can be formulated as the optimal power flow model. The loss minimization $L_{ij,t}^{\varphi} r_{ij}^{\varphi}$ and cost minimization $\lambda_{0,t} P_{0,t}^{\varphi}$ can be treated as the target objective, we will not list here due to the page limitation. Then the optimal power flow model results $\mathbf{x} = [P_{ij,t}, L_{ij,t}]$ will be transmitted to the carbon emission accounting procedure in the next subsection.

2) *Carbon Emission Flow in an Unbalanced Network*: In an unbalanced distribution network, CEF can trace carbon footprints on both the generation and consumption roles for a prosumer after obtaining power flow results. Therefore, the traditional CEF model [18] is extended to the unbalanced network to account for carbon emissions at a ‘nodal-phase’ level. The net phase level carbon injection $R_{j,t}^{inj,\varphi}$ is introduced as follows in equation (14):

$$R_{j,t}^{inj,\varphi} = e_{j(b)}^{netgen,\varphi(b)} P_{j(b),t}^{netgen,\varphi(b)} + e_j^{Res,\varphi} P_{j,t}^{Res,\varphi} + e_j^{MT,\varphi} P_{j,t}^{MT,\varphi}, \forall j, \varphi, t \quad (14)$$

The net carbon injection in (14) consists of net carbon emission generated by the prosumer as the term $e_{j(b)}^{netgen,\varphi(b)} P_{j(b),t}^{netgen,\varphi(b)}$ (i.e., the emission generated from the device belongs to the prosumer) and the RES, and the MT (i.e., not belong to the prosumer) in the distribution network. Note that only the generation-type prosumer can produce carbon emission injections. The carbon emission accounting for the consumption device depends on the emission flow accounting result to achieve $e_{j(b),t}^{\varphi(b)}$ in equation (15). The device carbon intensity for RES $e_j^{Res,\varphi}$ and MT $e_j^{MT,\varphi}$ are defined as the system input parameters and are fixed in general.

Before introducing the emission accounting, we first utilize results $\mathbf{x}$ from the optimal power flow model under constraint in (1)–(13). To be specific, the active powerflow $P_{ij,t}$ and squared line current $L_{ij,t}$ is utilized. Then the direction of the $P_{ij,t}$ needs to be identified, allowing to track the carbon flow direction in the power network.

We then introduce the new set $\mathcal{N}_j^+$ and $\mathcal{N}_j^-$ to describe the power flow direction in $P_{ij,t}$, as similar in [32]. It indicates the set of neighbor nodes that send power to (receive power from) node j , respectively. For example, if the active power flows into the node j , the quantity is written as the new variable $P_{mj,t}^{j,\varphi}$,

$\forall m \in \mathcal{N}_j^+$. On the other hand, if the active powerflow out of the node j , the quantity is written as the other new variable $P_{jn,t}^{j,\varphi}$, $\forall n \in \mathcal{N}_j^-$. According to the known power flow direction, new introduced variables $P_{mj,t}^{j,\varphi}$ and $P_{jn,t}^{j,\varphi}$ can be derived for emission accounting in the following equation (15).

The *proportional sharing principle* is applied as in [32] to achieve the carbon intensity. The carbon intensity $e_{j,t}^{\varphi}$ for node j at phase φ can be calculated as:

$$e_{j,t}^{\varphi} = \frac{R_{j,t}^{inj,\varphi} + \sum_{m \in \mathcal{N}_j^+} (e_{mj,t}^l P_{mj,t}^{j,\varphi} - L_{mj,t}^{\varphi} r_{mj}^{\varphi} e_{mj,t}^{l,\varphi})}{P_{j,t}^{gen,\varphi} + \sum_{m \in \mathcal{N}_j^+} (P_{mj,t}^{j,\varphi} - L_{mj,t}^{\varphi} r_{mj}^{\varphi})}, \quad (15)$$

$$\forall j \in \mathcal{N}, \forall \varphi, \forall m \in \mathcal{N}_j^+$$

$$e_{jn,t}^{l,\varphi} = e_{j,t}^{\varphi}, \forall j \in \mathcal{N}, \forall n \in \mathcal{N}_j^-, \forall \varphi, t \quad (16)$$

In equation (15), $e_{j,t}^{\varphi}$ denotes the ‘nodal-phase’ level carbon intensity, it is calculated from the net carbon emission injection at the numerator divided by the net generated active power injection at the node j at the denominator (i.e., the injected power from generation device $P_{j,t}^{gen,\varphi}$ and the power flow into the node j). Similar to the power line loss in the conventional powerflow calculation, carbon emission leakage cannot be neglected in the distribution network. The line loss leakage is written as the term $L_{mj,t}^{\varphi} r_{mj}^{\varphi} e_{mj,t}^{l,\varphi}$. The loss leakage is calculated with the general line loss multiplied by line carbon intensity $e_{mj,t}^{l,\varphi}$. The line carbon intensity $e_{jn,t}^{l,\varphi}$ of power outflow line $l = (j, n)$ written in (16) is determined by the intensity at the start node of the line l , in this case, node j is the start node and node n is the end node. Apparently, the carbon line intensity $e_{mj,t}^{l,\varphi}$ and node intensity $e_{j,t}^{\varphi}$ follows the recursive relationship, it can be solved iteratively to achieve the final solutions. With the accounted emission to each node, the carbon footprints can be traced from the slack bus to the end of the feeder. To this end, the emission from load type prosumer (e.g., demand side units) connected at φ node j can be accounted. For details, the multiplication with derived node-phase level intensity $e_{j(b),t}^{\varphi(b)}$ and the term load quantity $P_{j(b),t}^{load,\varphi(b)}$ subtract demand response quantity $P_{j(b),t}^{Dr,\varphi(b)}$.

The carbon emission for each node j with phase φ can be calculated with the carbon emission inflow from originating node and the emission generated at the receiving node. As a result to consider equation emission from originating and receiving node, the allocation coefficient is defined as 1/2 as in [11]. The carbon emission generated at the receiving node is calculated by summing the demand side and generation side.

Therefore, the ‘nodal-phase’ level of carbon emission $C_{j,t}^{\varphi}$ with prosumer injection and carbon inflow can be written as follows in equation (17):

$$C_{j,t}^{\varphi} = \frac{1}{2} \sum_{m \in \mathcal{N}_j^+} e_{mj,t}^l P_{mj,t}^{j,\varphi} + \frac{1}{2} \cdot \underbrace{\left[(P_{j,t}^{L,\varphi} e_{j,t}^{\varphi}) + (P_{j(b),t}^{load,\varphi(b)} - P_{j(b),t}^{Dr,\varphi(b)}) \cdot e_{j(b),t}^{\varphi(b)} \right]}_{\text{Demand side}} + \underbrace{e_j^{MT,\varphi} P_{j,t}^{MT,\varphi} + e_{j(b)}^{netgen,\varphi(b)} P_{j(b),t}^{netgen,\varphi(b)}}_{\text{Generatoin side}}, \forall j, \varphi, t \quad (17)$$

For each node j , the carbon emissions consider both the carbon inflow into this node from the neighboring node sets $e_{mj,t}^l P_{mj,t}^{j,\varphi}$, $m \in \mathcal{N}_j^+$, and the net carbon injection from the current node.

Consequently, with the carbon footprint tracing from the generation side to the load side at the system level, the emission for changeable roles of the prosumer can be efficiently allocated. Then, the system-wide ‘nodal-phase’ level carbon emission constraints is derived as:

$$C_{j,t}^\varphi \leq \bar{C}_{j,t}^\varphi + \hat{C}_{j,t}^\varphi \quad \forall j, \varphi, t \quad (18)$$

where $\bar{C}_{j,t}^\varphi$ denotes the ‘nodal-phase’ level initial carbon emission permits, $\hat{C}_{j,t}^\varphi$ indicates the excessive carbon emission permits, which are the additional emission permits that need to be purchased by DSO from the upstream network if the initial permits are used out to satisfy constraint (18). Lastly, the DSO can provide carbon-aware service prices based on the ‘nodal-phase’ level constraints in following the CAOPF model.

3) *Carbon Aware Market Clearing Model*: The CAOPF follows the procedure of pre-dispatch from getting the results of OPF, emission accounting and CAOPF market clearing. Specifically, using the allocated carbon intensity, the constraint in (18) can be seamlessly integrated into the OPF calculation process. Additionally, the carbon permit purchase cost $\lambda_{c,t} \hat{C}_{j,t}^\varphi$ from the wholesale market is incorporated into the objective function. This leads to the reformulation of the problem as the CAOPF for carbon-conscious market clearing in (19). The model is summarized as follows:

Objective:

$$\min \sum_{t \in T} \sum_{\varphi \in \Phi} \left(\sum_{j \in \mathcal{N}} (\lambda_{c,t} \hat{C}_{j,t}^\varphi) + \lambda_{0,t} P_{0,t}^\varphi + \sum_{(i,j) \in \mathcal{E}} L_{ij,t}^\varphi \right) \quad (19a)$$

Subject to:

$$\text{Active/reactive power balance constraints : (1) – (3)} \quad (19b)$$

$$\text{Voltage drop constraint : (7)} \quad (19c)$$

$$\text{Voltage limitation constraint : (12)} \quad (19d)$$

$$\text{SOCP constraint : (13)} \quad (19e)$$

$$\text{Carbon emission permits constraint : (18)} \quad (19f)$$

The objective function consists of the cost of total excessive ‘nodal-phase’ level carbon permits, the total power that needs to be purchased at the slack bus, and the network losses. The price $\lambda_{c,t}$ and $\lambda_{0,t}$ indicate the carbon excessive price and electricity prices at the slack bus. Then, we define the Lagrange multiplier $u_{j,t}^\varphi$, which is associated with the active power balance constraint in equation (1) as well as the first constraint in equation (19b) of the reformulated CAOPF model. Similarly, the $u_{j,t}^\varphi$ is associated with the constraint (19f).

The ‘nodal-phase’ level carbon price $\lambda_{j,t}^{c,\varphi}$ can be obtained through first-order derivative rule: we take the partial derivative to the Lagrange function with respect to the net injected power $P_{j,t}^{tie,\varphi}$ to obtain the equation (20a).

$$\frac{\partial L}{\partial P_{j,t}^{tie,\varphi}} = \lambda_{j,t}^\varphi, \quad \forall j, \varphi, t \quad (20a)$$

$$\frac{\partial L}{\partial C_{j,t}^\varphi} = u_{j,t}^\varphi, \quad \forall j, \varphi, t \quad (20b)$$

$$\frac{\partial L}{\partial \hat{C}_{j,t}^\varphi} = \lambda_{c,t} - \mu_{j,t}^\varphi, \quad \forall j, \varphi, t \quad (20c)$$

$\lambda_{j,t}^\varphi$ can be used for P2G electricity price as distribution location marginal price for electricity (DLMPE). Meanwhile, according to KKT condition, we let the inequality constraint (19f) equal to zero, it can be written as $\mu_{j,t}^\varphi (C_{j,t}^\varphi - \bar{C}_{j,t}^\varphi - \hat{C}_{j,t}^\varphi) = 0$. Then, we applied the same method to take the first-order derivative to the Lagrange function with respect to the carbon emissions $C_{j,t}^\varphi$ and excessive carbon emission $\hat{C}_{j,t}^\varphi$, achieving the equation (20b) and (20c). According to the complementary slackness in KKT condition, if the constraint (19f) is not active (i.e., $C_{j,t}^\varphi < \bar{C}_{j,t}^\varphi + \hat{C}_{j,t}^\varphi$), $u_{j,t}^\varphi$ equals to zero, it means that there is no additional carbon permits charge to node j phase φ . On the other hand, if the constraint (19f) is active (i.e., $C_{j,t}^\varphi = \bar{C}_{j,t}^\varphi + \hat{C}_{j,t}^\varphi$), and we can derive that $\mu_{j,t}^\varphi > 0$ and $\lambda_{c,t} - \mu_{j,t}^\varphi = 0$, the multiplier $u_{j,t}^\varphi$ equals to the additional carbon permits price $\lambda_{c,t}$. Therefore, extracting the Lagrange multiplier of this constraint (19f) can reflect the carbon emissions condition at this ‘nodal-phase’ level.

Furthermore, we aim to integrate the advantages of the previous methods [33] while introducing improvements in our approach. This is intended to further amplify the impact of carbon emissions on electricity pricing, thereby providing more incentive-driven signals for the lower-level P2P trading. Then the price is written as in equation (20c)

$$\lambda_{j,t}^{c,\phi} = \frac{u_{j,t}^\phi}{2} \cdot e_{j,t}^{\phi,co}, \quad \forall j, \varphi, t \quad (20d)$$

$$e_{j,t}^{\phi,co} = \frac{e_{j,t}^\phi}{e_{ref}^\phi}, \quad \forall j, \varphi, t \quad (20e)$$

In equation (20d), the extracted multiplier $u_{j,t}^\phi$ is multiplied by the carbon response coefficient term $e_{j,t}^{\phi,co}$. It can be calculated based on the equation (20e), the e_{ref}^ϕ is the unit carbon intensity to normalize the $e_{j,t}^\phi$. The intuition is to emphasize the carbon intensity for each node on the carbon price, $e_{j,t}^{\phi,co}$ is treated as a dimensionless scalar in this calculation as inspired from [34]. The factor of 1/2 in the denominator denotes the allocation ratio in equation (17), ensuring a balanced attribution of the nodal impact on carbon emissions or power flow. This approach allows us to capture its influence more effectively for the carbon intensity to the carbon price. The core idea is to externalize the emission charge explicitly and provide a service price for the P2P market to incentivize carbon-aware trading, i.e., distribution locational marginal price for carbon (DLMPC). To this end, the decomposed price for electricity $\lambda_{j,t}^\varphi$ (\$/kWh) and price for carbon permits $\lambda_{j,t}^{c,\varphi}$ (\$/tonsCO₂) has been successfully derived to the ‘nodal-phase’ level.

B. Prosumer Operation Model

The prosumer model at lower level for each prosumer b connected at node j phase φ and time t in (18) as follows:

$$\text{Obj: } \min \sum_{t \in T} [C(P2P) + C(P2G) + C(U)] \quad (21a)$$

$$\text{s.t. } p_{j(b),t}^{\text{net},\varphi(b)} = p_{b,t}^{\text{load}} - p_{b,t}^{\text{Dr}} - p_{b,t}^{\text{Res}} - p_{b,t}^{\text{MT}} \quad (21b)$$

$$c_{b,t}^{\text{pr}} = \frac{1}{2} \left(p_{b,t}^{\text{load},\varphi} - p_{b,t}^{\text{Dr}} \right) \cdot e_{j(b),t}^{\varphi(b)} + \frac{1}{2} \left(e_{j(b)}^{\text{netgen},\varphi(b)} p_{j(b),t}^{\text{netgen},\varphi(b)} \right) \quad (21c)$$

$$0 \leq p_{b,t}^{\text{MT}} \leq \bar{p}_{b,t}^{\text{MT}}, 0 \leq p_{b,t}^{\text{Dr}} \leq \varepsilon_b^{\text{dr}} p_{b,t}^{\text{load}} \quad (21d)$$

$$C(U) = \alpha_b^{\text{dr}} \left(p_{b,t}^{\text{load},\varphi} - p_{b,t}^{\text{Dr}} \right)^2 + \alpha_b^{\text{MT}} \left(p_{b,t}^{\text{MT}} \right)^2 + \beta_b^{\text{MT}} p_{b,t}^{\text{MT}} \quad (21e)$$

$$C(P2P) = \sum_{m_1} \lambda_{m_1,b,t}^e \cdot p_{m_1,b,t}^{\text{p2pe}} \cdot \mathbb{K}_{b,t}^e + \sum_{m_2} \lambda_{m_2,b,t}^c \cdot c_{m_2,b,t}^{\text{p2pc}} \cdot \mathbb{K}_{b,t}^c \quad (21f)$$

$$C(P2G) = \lambda_{j(b),t}^{\varphi(b)} \left[p_{b,t}^{\text{p2g}} \right]^+ + \lambda_t^{\text{FiT}} \left[p_{b,t}^{\text{p2g}} \right]^- + \lambda_{j(b),t}^{\text{c},\varphi(b)} \left[c_{b,t}^{\text{p2g}} \right]^+ \quad (21g)$$

$$p_{b,t}^{\text{bid}} = p_{j(b),t}^{\text{net},\varphi(b)} \quad (21h)$$

$$c_{b,t}^{\text{bid}} = c_{b,t}^{\text{pr}} - c_{b,t}^{\text{cap}} \quad (21i)$$

The objective consists of the P2P trading cost $C(P2P)$, the P2G trading cost $C(P2G)$ and the internal operation cost $C(U)$. Equation (21b) denotes the power balance, where the netload, $p_{j(b),t}^{\text{net},\varphi(b)}$ can be decided upon the demand response quantity $p_{b,t}^{\text{Dr}}$, RES output for the prosumer $p_{b,t}^{\text{Res}}$, the MT output for the prosumer $p_{b,t}^{\text{MT}}$ are decided. The netload is considered the bid value in equation (21h), it will be submitted to the P2P market for market clearing. The sign of the netload determines the role of buyer or seller in the auction market. Meanwhile, the equation (21c) denotes the net carbon emission produced by prosumer b . The equation (21d) denotes the output power range for $p_{b,t}^{\text{MT}}$ and demand response $p_{b,t}^{\text{Dr}}$. Where $\bar{p}_{b,t}^{\text{MT}}$ denotes the maximum output and $\varepsilon_b^{\text{dr}}$ indicates the load flexibility ratio.

As similar to the emission accounting in DSO level in equation (17), the prosumer emission accounting also follows the rule that 1/2 is used as the allocation ratio. The demand side carbon intensity $e_{j(b),t}^{\varphi(b)}$ is traced by the DSO in equation (17) and the generation side carbon intensity $e_{j(b)}^{\text{netgen},\varphi(b)}$ is determined locally through the proportional rules. Unlike the trading rules in the electricity market, the buyer and seller can be determined by the positive and negative signs of the calculated netload. The surplus and deficit carbon emission permits for bidding are determined by the difference between the net carbon emission $c_{b,t}^{\text{pr}}$ and allowable carbon permits $c_{b,t}^{\text{cap}}$ for each prosumer. A similar carbon market settlement can also be found in [35], [36]. The utility function in (21e) includes the DR cost function for user discomfort and MT operation cost. The α_b^{dr} , α_b^{MT} and β_b^{MT} are their corresponding cost coefficients.

The P2P cost in the JECT market is given in (21f), where m_1/m_2 is the index of matched peers for electricity/carbon trading. M_1/M_2 denotes the corresponding maximum number of matched transaction pairs. $\lambda_{m_1,b,t}^e$ and $\lambda_{m_2,b,t}^c$ denote the

Algorithm 1 Joint Energy and Carbon CDA Market Clearing Procedure

Input: Prosumers' dispatch intention $p_{b,t}^{\text{MT}}, p_{b,t}^{\text{Dr}}$, bidding price intention $\lambda_{b,t}^{\text{bid},e}, \lambda_{b,t}^{\text{bid},c}$, load and RES's profile $p_{b,t}^{\text{load}}, p_{b,t}^{\text{Res}}$
Output: Cleared E&C P2P trading price $\lambda_{b,t}^{\text{e}}, \lambda_{b,t}^{\text{c}}$, P2P trading quantity $p_{b,t}^{\text{p2pe}}, c_{b,t}^{\text{p2pc}}$, and P2G trading quantity $p_{b,t}^{\text{p2g}}, c_{b,t}^{\text{p2g}}$

For auction period (1h) $t = 1$ to T **do**

For P2P prosumer $b = 1$ to M **do**

1 Calculate the netload $p_{j(b),t}^{\text{net},\varphi(b)}$ in Eq. (21b) and net carbon emission $c_{b,t}^{\text{pr}}$ in Eq.(21c)

2 Calculate the electricity/carbon permit bidding quantity $p_{b,t}^{\text{bid},e}/c_{b,t}^{\text{bid}}$ in (21h), (21i) respectively

End for

3 Prosumers submit their electricity quantity and price bids ($p_{b,t}^{\text{bid},e}, \lambda_{b,t}^{\text{bid},e}$), carbon permit quantity and price bids ($c_{b,t}^{\text{bid}}, \lambda_{b,t}^{\text{bid},c}$) to the auctioneer (P2P platform operator)

For round (15min) $r = 1$ to R **do**

4 Auctioneer collects energy buyer orderbook $k_{t,r}^{\text{e,buy}} = \{(p_{\beta^e,t}^{\text{bid}}, \lambda_{\beta^e,t}^{\text{bid},e}) | p_{\beta^e,t}^{\text{bid}} > 0\}$ in descending order energy seller orderbook $k_{t,r}^{\text{e,sell}} = \{(p_{\kappa^e,t}^{\text{bid}}, \lambda_{\kappa^e,t}^{\text{bid},e}) | p_{\kappa^e,t}^{\text{bid}} < 0\}$ in ascending order

5 Auctioneer collects carbon permit buyer orderbook $k_{t,r}^{\text{c,buy}} = \{(c_{\beta^c,t}^{\text{bid}}, \lambda_{\beta^c,t}^{\text{bid},c}) | p_{\beta^c,t}^{\text{bid}} > 0\}$ in descending order carbon permit seller orderbook $k_{t,r}^{\text{c,sell}} = \{(c_{\kappa^c,t}^{\text{bid}}, \lambda_{\kappa^c,t}^{\text{bid},c}) | p_{\kappa^c,t}^{\text{bid}} < 0\}$ in ascending order

6 Auctioneer conducts electricity **DA market clearing** ($k_{t,r}^{\text{e,buy}}, k_{t,r}^{\text{e,sell}}$)

7 Auctioneer conducts carbon emission permits **DA market clearing** ($k_{t,r}^{\text{c,buy}}, k_{t,r}^{\text{c,sell}}$)

8 Prosumers update their price bids ($\lambda_{\beta^e,t}^{\text{bid},e}, \lambda_{\kappa^c,t}^{\text{bid},c}$) according to the updated orderbook

End for

9 Auctioneer publishes electricity trading outcomes, including P2P trading price $\lambda_{b,t}^{\text{e}}$, P2P trading quantity $p_{b,t}^{\text{p2pe}}$, unmatched quantity $p_{b,t}^{\text{p2g}}$

10 Auctioneer publishes carbon trading outcomes, including P2P trading price $\lambda_{b,t}^{\text{c}}$, P2P trading quantity $c_{b,t}^{\text{p2pc}}$, unmatched quantity $c_{b,t}^{\text{p2g}}$

11 Balance the unmatched energy quantity $p_{b,t}^{\text{p2g}}$ with DLMPE, and deficit carbon permit $c_{b,t}^{\text{p2g}}$ with DLMPC in (21g)

End for

Function: DA market clearing

Input: buyer and seller orderbook at time step t , round r : $k_{t,r}^{\text{e,buy}}, k_{t,r}^{\text{e,sell}}$

Output: matched quantity-price ($\lambda_{t,r}^m, \sigma_{t,r}^m$), unmatched quantity $\sigma_{b,t}^{\text{p2g}}$

Initialize indices of buyer and seller $\beta = \kappa = 1$

Initialize the number of matched pairs $m = 1$

While $\lambda_{\beta,t,r} \geq \lambda_{\kappa,t,r}$ (social welfare exists) **do**

Match the trading quantity $\sigma_{t,r}^m = \min(\sigma_{\beta,t,r}, \sigma_{\kappa,t,r})$

Calculate the trading price $\lambda_{t,r}^m = (\lambda_{\beta,t,r} + \lambda_{\kappa,t,r})/2$

Update buyer orderbook $\sigma_{\beta,t,r} \leftarrow \sigma_{\beta,t,r} - \sigma_{t,r}^m$ and seller order book $\sigma_{\kappa,t,r} \leftarrow \sigma_{\kappa,t,r} - \sigma_{t,r}^m$
 $m \leftarrow m + 1$

If $q_{\beta,t,r} = 0$ **then** $\beta \leftarrow \beta + 1$

If $q_{\kappa,t,r} = 0$ **then** $\kappa \leftarrow \kappa + 1$

Break if $\beta > \text{length of buyer orderbook } N_\beta$ or $\kappa > \text{length of seller orderbook } N_\kappa$

End while

Bidding quantities remain in the buyer and seller orderbook are unmatched $\sigma_{b,t}^{\text{p2g}}$

cleared P2P electricity and carbon price from the auctioneer, which are derived from the CDA market clearing procedure (details are provided in Algorithm 1). Meanwhile, $p_{m_1,b,t}^{\text{p2p}}$ and $c_{m_2,b,t}^{\text{p2pc}}$ denote the cleared P2P electricity and carbon permits trading quantity. $\mathbb{K}_{b,t}^e$ and $\mathbb{K}_{b,t}^c$ denote the trading role of the prosumer, where $\mathbb{K}_{b,t} = 1$ indicates it is the buyer and $\mathbb{K}_{b,t} = -1$ is the seller. Note that the buying and selling roles

of prosumers in electricity trading and carbon trading are independent of each other. In the JECT market, a prosumer can simultaneously act as an electricity seller and a carbon permits buyer, and vice versa. This significantly enhances market flexibility, further increasing the social welfare. Lastly, the unmatched quantity in JECT market will be cleared in the P2G market with DSO, the payment is written in (21g). It should be noted that the prosumer can sell unmatched electricity power to the DSO with a feed-in tariff (FiT) λ_t^{FiT} , but it is not allowed for prosumers to sell the carbon emission permits back to the DSO.

C. Carbon-Aware P2P Trading Market Model

With the price obtained from DSO, the profit gap incentivizes the prosumer-based agent for peer matching in the JECT P2P market. The CDA market is a highly efficient market mechanism. It can match multiple prosumers (with changeable roles as buyers and sellers) who participate in the P2P trading platform. It has been widely adopted in the electricity P2P trading market. For this paper, we proposed the JECT CDA market.

The core idea of the CDA is to provide incentives for as much quantity as cleared at the P2P market. After the electricity $p_{b,t}^{bid}$ and carbon emission bid value $c_{b,t}^{bid}$ is determined. They can submit the bidding price decisions to the auctioneer. To be specific, the bidding price should be maintained in the following range:

$$\lambda_t^{FiT} \leq \lambda_{b,t}^{bid,e} \leq \lambda_{j(b),t}^{mean,\varphi(b)} \quad (22)$$

$$0 \leq \lambda_{b,t}^{bid,c} \leq \lambda_{j(b),t}^{mean,c,\varphi(b)} \quad (23)$$

Equation (22) indicates the bidding price between Fit λ_t^{FiT} and the mean DLMPE. With the same setting in equation (23), the bidding price for selling and buying in the carbon emission market should be maintained at the range between zero (cannot sell back to the DSO) and mean DLMPC. The reason for constraining the bidding range is to emphasize the practical economic rationality within the JECT P2P trading market. Using the role of buyer as an example, no buyer would willingly purchase electricity at a price higher than the retailer's price (i.e., the upper bound price in (22)–(23)), as they can get cheaper electricity directly from the retailer. On the other hand, no seller would sell electricity at a price lower than FiT (i.e., the lower bound price in (22)–(23)), as they could sell directly to the grid for a higher return. In real-world scenarios, it is common for peers to be aware of the upper and lower bounds of bidding prices. The most classic case is the use of retail prices and feed-in tariffs, where P2P trading typically operates within this price range. At the same time, to ensure market clearing stability and prevent malicious bidding, we used the average price $[\lambda_{j(b),t}^{mean,\varphi(b)}, \lambda_{j(b),t}^{mean,c,\varphi(b)}]$ as the basis for defining the bidding range. It should be noted that while the bidding range is based on the average value, the final market clearing for peer-to-grid trading with DSO is still based on the actual value $[\lambda_{j(b),t}^{\varphi(b)}, \lambda_{j(b),t}^{c,\varphi(b)}]$.

There are four main procedures for the JECT bidding in **Algorithm 1**.

Prosumer Bidding Preparation: Each prosumer b calculates the netload $p_{j(b),t}^{net,\varphi(b)}$ in (21b) and net carbon emission $c_{b,t}^{pr}$ in (21c) in Step 1. Then, the electricity/carbon permit trading quantity can be determined in (21h) and (21i) in Step 2.

Bidding and Auction Execution: All prosumers submit their JECT bidding quantities and prices. Note that the quantity bid is derived from Steps 1 and 2, and the price bid is derived from the action space output in (25). Then, the auctioneer collects their JECT order book for the buyer and seller (i.e., the role depends on the sign of quantity bid in Step 1). In Steps 6 and 7, the auctioneer conducts the double auction (DA) algorithm. The details of DA are listed in **Algorithm 1** below as the function call. In each trading round r , the trading outcome includes the matched pair quantity m , the cleared quantity price for each m , and unmatched quantity. In Step 8, the prosumer can update their price bids according to the prior bidding round $r-1$. In the reinforcement learning scheme, the process is adaptive, and they can learn from their success and failure in the previous bids. Considering the varied flexibility of each prosumer, extensively adjusting the quantity bid is impractical. Therefore, in our case, we only update the price bid $(\lambda_{\beta,t}^{bid,e}, \lambda_{\kappa,t}^{bid,c})$.

Clearing Results: When the auction round reaches the max round R , the auctioneer publishes the final trading outcome for the JECT market in Steps 9 and 10.

Balancing and Settlement: For the unmatched quantity $p_{b,t}^{p2g}$ and $c_{b,t}^{p2g}$, the prosumer needs to trade with the DSO with the DLMPE $\lambda_{j(b),t}^{\varphi(b)}$ and DLMPC $\lambda_{j(b),t}^{ca,\varphi(b)}$ in (21g) if they want to purchase quantity from the DSO. On the other hand, they need to sell quantity to DSO; Fit is used for electricity trading, and zero is used for carbon trading.

Under this designed market clearing mechanism, prosumers can quickly match based on their bids to maximize social welfare ($\lambda_{\beta,t,r} \geq \lambda_{\kappa,t,r}$). Our motivation for designing the market clearing process is to minimize the prosumer's operating model defined by (21a)–(21i) through JECT P2P matching.

Remark 1: The DSO price $\lambda_{j(b),t}^{\varphi(b)}, \lambda_{j(b),t}^{c,\varphi(b)}$ are always uncertain and unknown before the interaction, and it is difficult to predict, as proved in [17]. Given the price fluctuations, prosumers may face reduced profits compared to a more conservative bidding strategy in the joint CDA market. In our case, historical data can be used as the initial start value for training.

IV. NASH DISCOVERY MADRL

Due to the highly stochastic and dynamic nature of the joint CDA market, solving an optimization problem for quantity bids requires detailed knowledge of both the prosumer and DSO operational models. Such a requirement is typically impractical in real-world scenarios. Thus, implementing a cooperative Markov Game becomes a more feasible approach. Additionally, the NE serves as a crucial performance metric to assess the stability and convergence of bidding strategies in a multi-agent system. This section explores the NE-discovery¹

¹NE-discovery refers to approach the NE solution during the MADRL training progress.

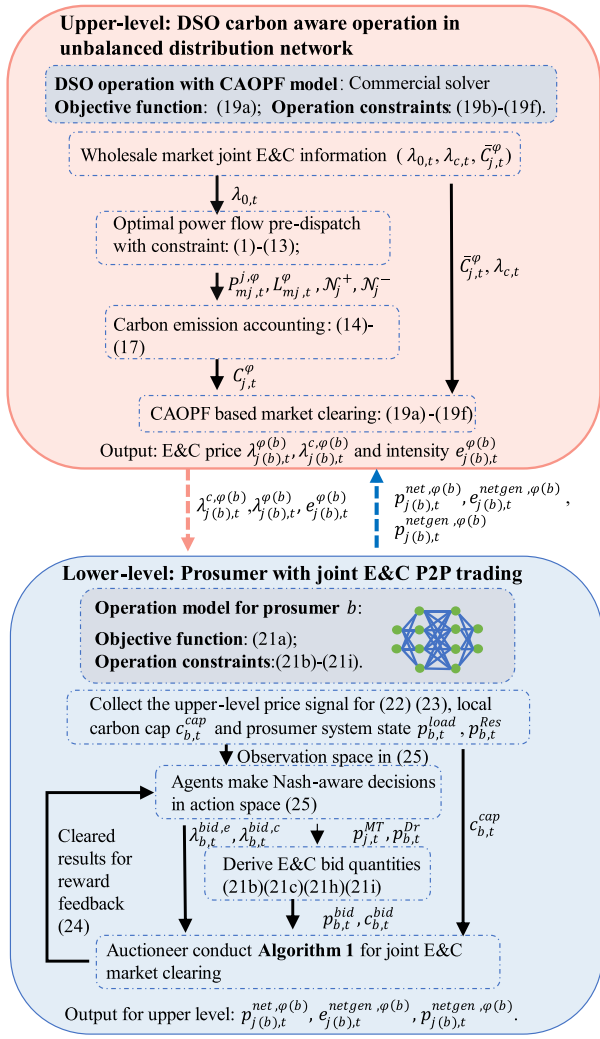


Fig. 2. The bilevel interaction framework.

MADRL method based on the trust region policy optimization (TRPO) algorithm.

A. Bilevel Interaction Model

Upper level: The upper level is the DSO agent in Fig. 2. The DSO operation model is based on the CAOPF and can be solved by commercial optimizer, the objective function is to optimize the cost in equation (19a) while satisfying the constraint (19b)–(19f). However, the CAOPF cannot be solved directly since the carbon emission accounting process (14)–(17) relies on the OPF results under the constraints (1)–(13). The arrows in the diagram represent the information flow of variables, ultimately leading to the final CAOPF solution process. The CAOPF model at the upper-level omits the output of DLMPC $\lambda_{j(b),t}^{c,\varphi(b)}$ and DLMPE $\lambda_{j(b),t}^{\varphi(b)}$ to the lower level.

Lower level: The lower level is modeled by DRL agents for prosumers engaged in JECT P2P trading in Fig. 2. The model's objective function is defined in (21a), while its operational constraints are outlined in (21b)–(21i). Prosumers receive price signals $(\lambda_{j(b),t}^{\varphi(b)}, \lambda_{j(b),t}^{c,\varphi(b)})$ from the upper level as defined in (22) and (23), along with the local carbon

cap $c_{b,t}^{cap}$ and system states $(p_{b,t}^{load}, p_{b,t}^{res})$. Using this as the input, the observation space is constructed as described in (25), enabling agents to make Nash-aware decisions within the action space, balancing electricity and carbon considerations. Based on these decisions, prosumers derive E&C bid quantities by following equations (21b), (21c), (21h), and (21i). The auctioneer then conducts **Algorithm 1** for joint E&C market clearing, producing market clearing results. These results are fed back into reward calculation as described in (24), enabling policy improvement for training in the environment.

Coupling variables between two levels: The upper level transmits the DLMPC $\lambda_{j(b),t}^{c,\varphi(b)}$, DLMPE $\lambda_{j(b),t}^{\varphi(b)}$ and carbon intensity $e_{j(b),t}^{\varphi(b)}$ to the lower level. The lower level calculates the net generation power injection $P_{j(b),t}^{netgen,\varphi(b)}$, net carbon generation intensity $e_{j(b),t}^{netgen,\varphi(b)}$ and net power injection $P_{j(b),t}^{net,\varphi(b)}$. In the actual interaction process of reinforcement learning, the lower-level prosumers are treated as agents, while the upper level is regarded as part of the environment. This creates a dynamic interaction between the two layers within the environment. During training, the lower-level agents collect data through interactions with the environment and improve their policies accordingly.

B. Cooperative Markov Game Formulation

The cooperative Markov game can be defined as the following tuple $(M, \mathcal{S}, \mathcal{A}, R, \mathcal{P}, \gamma)$, where M denotes the number of agents. $\mathcal{S}$ is the system state space, and $\mathcal{A} = \prod_{b=1}^M \mathcal{A}^b$ the joint action space. Then, the joint reward function can be written as $\mathcal{S} \times \mathcal{A} \rightarrow R$. The $\mathcal{S}$ refers to the state space. The transition probability can be defined as $\mathcal{P}: \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow [0, 1]$, which is unknown before the training. γ denotes the discount factor, which can be written as $\gamma \in [0, 1]$. The joint actions are derived as $\mathbf{a}_t = (a_t^1, \dots, a_t^M)$, $\forall t \in \mathcal{T}$ from their joint policy $\pi(\cdot|s_t) = \prod_{b=1}^M \pi^b(\cdot|s_t) \in \mathcal{P}(\mathcal{A})$ which guides the agents' joint decisions under the system state s_t .

1) **Reward:** The Reward for $\forall t \in T$ Can Be Written as Follows

$$r_{b,t} = -[C(P2P) + C(P2G) + C(U)] \quad (24)$$

The reward in (24) indicates the negative cost summarization for in the P2P2G market as derived in (21a) for agent b . The joint reward R_t can be calculated as: $\sum_{b=1}^M r_{b,t}$.

2) **Action Space:** The action space includes the prosumer unit as follows:

$$\mathcal{A}_t^b = [p_{b,t}^{MT}, p_{b,t}^{Dr}, \lambda_{b,t}^{bid,e}, \lambda_{b,t}^{bid,c}] \in \mathcal{A} \quad (25)$$

The action space includes MTs' dispatch signal, and the DR signal. The action range for $p_{b,t}^{MT}$ and $p_{b,t}^{Dr}$ is constrained in (21d). $\lambda_{b,t}^{bid,e}, \lambda_{b,t}^{bid,c}$ indicate the bidding price for the JECT market, and the range is constrained in (22) and (23).

3) **Observation Space:** The observation space for agent b is designed as follows:

$$o_t^b = \left[e_{j(b),t}^{\varphi(b)}, e_{j(b)}^{\text{netgen},\varphi(b)}, p_{b,t}^{\text{load}}, p_{b,t}^{\text{Res}}, p_{b,t-1}^{\text{bid}}, c_{b,t-1}^{\text{bid}}, \lambda_{j(b),t-1}^{\varphi(b)}, \lambda_{j(b),t-1}^{c,\varphi(b)} \right] \in \mathcal{S} \quad (26)$$

The observation space consists of phase-level carbon intensity $e_{j(b),t}^{\varphi(b)}$ for node j and phase φ attached with agent b , the generation intensity $e_{j(b)}^{\text{netgen},\varphi(b)}$, the load consumption, and the renewable output profile. Meanwhile, the bidding strategy from the previous time step is also observed for learning efficiency. Lastly, the price cleared by DSO from the previous time step $t-1$ are recorded for observation to adjust the bidding strategy for the current time slot.

C. Preliminary for Multi-Agent Policy Learning

The preliminaries contain the policy optimization target $J(\pi)$ and how we can derive the advantage function $A_\pi(s, \mathbf{a})$ for joint actions $\mathbf{a}$, allowing target maximization. We first present the policy optimization target $J(\pi)$.

$$J(\pi) \triangleq \mathbb{E}_{\mathbf{a}_{0:T} \sim \pi} \left[\sum_{t=0}^T \gamma^t R_t \right] \quad (27)$$

In equation (27), the optimization target J is the function of joint policy π , the joint policy π omits the joint actions $\mathbf{a}$ from the initial period 0 to the end of the step T . The return is calculated by the summarized joint reward R_t over this episode (i.e., from step 0 to step T). Intuitively, we aim to find the optimal joint policy π_* for all prosumers to maximize the joint return in (27).

Then we further analysis how the prosumer-based agents can take action to maximize the joint return. The action output depends on the output of the policy network by using the system state as the input $\mathbf{a} \sim \pi(\mathbf{a} | s)$. Therefore, we can shift from finding the optimal policy π_* to finding the optimal joint actions $\mathbf{a}$. To this end, we introduce the state action value function $Q_\pi(s, \mathbf{a})$ in equation (28).

$$Q_\pi(s, \mathbf{a}) \triangleq \mathbb{E}_{s_{1:T} \sim P, \mathbf{a}_{1:T} \sim \pi} \left[\sum_{t=0}^T \gamma^t R_t \mid s_0 = s, \mathbf{a}_0 = \mathbf{a} \right] \quad (28)$$

Different from the off-policy method that maximize the $Q_\pi(s, \mathbf{a})$ directly to find the optimal joint actions. We further introduce the joint advantage function $A_\pi(s, \mathbf{a})$ in equation (29), it can be derived as the difference between the $Q_\pi(s, \mathbf{a})$ and state value function $V_\pi(s)$ in (30). It can further quantify the improvement for particular joint actions $\mathbf{a}$ compared to the average performance of the policy in state s . Thus, it can guide the learning process to focus on actions that provide higher benefits relative to the baseline performance of the policy.

$$A_\pi(s, \mathbf{a}) \triangleq Q_\pi(s, \mathbf{a}) - V_\pi(s) \quad (29)$$

$$V_\pi(s) \triangleq \mathbb{E}_{s_{1:T} \sim P, \mathbf{a}_{1:T} \sim \pi} \left[\sum_{t=0}^T \gamma^t R_t \mid s_0 = s \right] \quad (30)$$

D. Multi-Agent Advantage Function Decomposition

We then provide definitions to examine the NE within the context of the aforementioned cooperative Markov game. NE

is a well-established concept in game theory, representing an agent's optimal bidding decision in response to the strategies of other agents in the coordination game. Next, we formally define the NE in the Markov game.

Definition 1: In a Markov game, a joint policy $\pi_* = (\pi_*^1, \dots, \pi_*^M)$ is the Nash equilibrium if it for every agent $b \in \Omega_{\text{prosumer}}$, $\pi^b \in \Pi^b$ implies $J(\pi_*) \geq J(\pi^b, \pi_*^{-b})$.

This definition characterizes the equilibrium point at convergence. In a multi-agent bidding problem, stability is evaluated by ensuring no agent is incentivized to change their policy. To analyze the NE of this coordination, the focus shifts to each agent's individual contribution to the joint reward and how they respond to others' strategies. The process of cooperative decision-making to maximize expected returns is further discussed.

Definition 2: Let the index term $b_{1:m}$ denote the ordered subset $\{b_1, \dots, b_m\}$ of the agent set Ω_{prosumer} . The term $-b_{1:m}$ indicates the rest of the agents in this set, the maximum number of agents is denoted as M . Then, we define b_m as the m th agent in this ordered subset. Therefore, the multi-agent state action function can be written as:

$$Q_\pi^{b_{1:m}}(s, \mathbf{a}^{b_{1:m}}) \triangleq \mathbb{E}_{\mathbf{a}^{-b_{1:m}}, \pi^{-b_{1:m}}} [Q_\pi(s, \mathbf{a}^{b_{1:m}}, \mathbf{a}^{-b_{1:m}})] \quad (31)$$

To be specific, if the index term $b_{1:m}$ permutations reach the maximum number of agents M in the Markov game, $Q_\pi^{b_{1:M}}(s, \mathbf{a}^{b_{1:M}})$ is equivalent to $Q_\pi(s, \mathbf{a})$. Then, considering two disjoint subset of agents $j_{1:k}$ and $b_{1:m}$, the multi-agent advantage function of agents $j_{1:k}$ with respect to $b_{1:m}$ is defined as:

$$A_\pi^{b_{1:m}}(s, \mathbf{a}^{j_{1:k}}, \mathbf{a}^{b_{1:m}}) \triangleq Q_\pi^{j_{1:k}, b_{1:m}}(s, \mathbf{a}^{j_{1:k}}, \mathbf{a}^{b_{1:m}}) - Q_\pi^{j_{1:k}}(s, \mathbf{a}^{j_{1:k}}) \quad (32)$$

In (32), the function $Q_\pi^{b_{1:M}}(s, \mathbf{a}^{b_{1:M}})$ evaluates the value of agents $b_{1:M}$ under the system state s considering the action in $-b_{1:M}$. Thus, the advantage function $A_\pi^{b_{1:m}}(s, \mathbf{a}^{j_{1:k}}, \mathbf{a}^{b_{1:m}})$ evaluates the advantages of taking action $\mathbf{a}^{j_{1:k}}$ based on the action taken by $\mathbf{a}^{b_{1:m}}$. However, it is still difficult to evaluate single agent's unique contribution under joint evaluation function. Intuitively, we need to decompose the joint advantage function in (31) for each agent b_m considering the bidding action from $-b_{1:m}$. Inspired by the theoretical finding in [37], the **Lemma 1** is introduced as follows.

Lemma 1 (Multi-Agent Advantage Decomposition): For any cooperative Markov games under the joint policy π and system state s with any agent subset $b_{1:M}$, the advantage function can be decomposed as:

$$A_\pi^{b_{1:M}}(s, \mathbf{a}^{b_{1:M}}) = \sum_{m=1}^M A_\pi^m(s, \mathbf{a}^{b_{1:m-1}}, \mathbf{a}^{b_m}) \quad (33)$$

Proof:

$$\begin{aligned} A_{\pi_\theta}^{b_{1:M}}(s, \mathbf{a}^{b_{1:M}}) &= Q_{\pi_\theta}^{b_{1:M}}(s, \mathbf{a}^{b_{1:M}}) - V_{\pi_\theta}(s) \\ &= \sum_{m=1}^M [Q_{\pi_\theta}^{b_{1:m}}(s, \mathbf{a}^{b_{1:m}}) - Q_{\pi_\theta}^{b_{1:m-1}}(s, \mathbf{a}^{b_{1:m-1}})] \\ &= \sum_{m=1}^M A_{\pi_\theta}^m(s, \mathbf{a}^{b_{1:m-1}}, \mathbf{a}^{b_m}) \end{aligned} \quad (34)$$

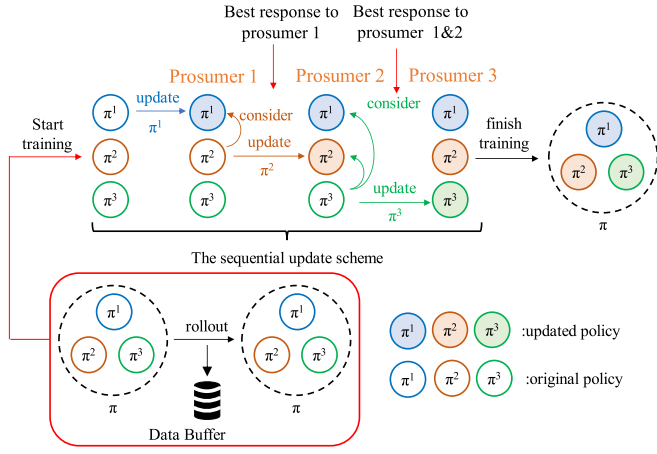


Fig. 3. Multi-agent decomposition and sequential update (3-agents example).

Lemma 1 provides an intuition for joint advantage function decomposition under rigorous proof. For the joint action $\mathbf{a}$, the lemma confirms that the advantage value, $A_{\pi_\theta}(s, \mathbf{a})$, is positive, ensuring the update performance is guaranteed, and it can also evaluate the action a^{bm} considering the update from the rest of the agents' actions $\mathbf{a}^{b_{1:m-1}}$. More details for discussion and analysis for Lemmal can be found in the supplementary document [38]. This decomposition method can be implemented by the sequential update, as shown in Fig. 3, with three prosumers as the simple 3-agent example.

In Fig. 3, the first part (referred to as the “upper” section in the diagram) decomposes the total advantage of a task into stages, taking into account the actions of previous agents. The second part (the “lower” section in the diagram) updates each agent’s policy during training by considering the earlier updates. As we can observe, the agent-based prosumer can provide the best response to the rest of the agents for each update. It leverages a foundation to find NE in the training game. It should also be noted that the update order is arbitrary for each round.

One might wonder why the order of agents in updates is randomized at each iteration. This condition is essential for ensuring convergence to an NE, which makes an intuitive sense: the fixed-point joint policies resulting from this randomized process guarantee that no agent is incentivized to make any further updates on the trading strategies, thus achieving an NE discovery. With this foundation, we further introduce a training framework under **Definition 2** and **Lemma 1** to prove that the learning can converge to NE during the training.

E. Training Framework for NE Discovery

The motivation of this subsection is to provide a general training framework for NE discovery rather than a specific algorithm. Inspired by the mirror learning (ML) framework [27], the NE can be proved if the following properties can be designed and **Lemma 1** is satisfied. First, we clarify the notation in this framework, using $\hat{\pi}^b$ to indicate the candidate update policy, π^b is the ready to update policy, and $\bar{\pi}^{j_{1:m}}$ is the updated policy in the learning process, given that agents $j_{1:m}$

just update to joint policy $\bar{\pi}^{j_{1:m}}$. There are three theoretical properties defined in [39] in this framework:

1. *Drift function*: Denoted by $\mathfrak{D}_\pi^b(\hat{\pi}^b | s, \bar{\pi}^{j_{1:m}})$. It indicates the distance between the ready to update policy π^b , and candidate policy $\hat{\pi}^b$. This function can be used to softly constrain the learning steps in a coordinated way under the NE approaching procedure.
2. *Neighboring operator*: The neighboring operator $\mathcal{U}^b$ is used for policy updates to produce small policy space subsets, resulting in a small and improved policy update. It is a desirable property when optimizing neural network policies, as it stabilizes the algorithm.
3. *Policy optimization target*:

$$\left[\mathcal{M}_{\mathfrak{D}_\pi^b, \bar{\pi}^{j_{1:m}}}^{(\hat{\pi}^b)} A_\pi \right](s) \triangleq \mathbb{E}_{\mathbf{a}^{j_{1:m}} \sim \bar{\pi}^{j_{1:m}}, \mathbf{a}^b \sim \hat{\pi}^b} \left[A_\pi^b(s, \mathbf{a}^{j_{1:m}}, \mathbf{a}^b) \right] - \mathfrak{D}_\pi^b(\hat{\pi}^b | s, \bar{\pi}^{j_{1:m}}) \quad (35)$$

The optimization target includes the advantage function as introduced in **Definition 2**, and the $\mathcal{M}$ denotes the learning operator to decompose the joint advantage function. Meanwhile, the target consists of the drift function $\mathfrak{D}^b$ to constrain the learning step as the penalty term. Note that mirror learning is a general framework for NE discovery. These theoretical properties need to be well designed for a specific environment, more details for mirror learning can be found in the supplementary document [38].

With these theoretical properties, we begin the proof of NE discovery under the ML framework. Three properties must be verified to guarantee the trained policy converge to the NE.

1. The value function can converge to the Nash value function V^{NE} ,

$$\lim_{k \rightarrow \infty} V_{\pi_k} = V^{NE}$$

2. Their expected returns converge to a Nash return,

$$\lim_{k \rightarrow \infty} J(\pi_k) = J^{NE}$$

3. Their ω -limit policy set consists of Nash equilibria.

Proof:

Step 1 (Convergence to the value function): With **Definition 2** and the unified ML framework [39], we have $V_{\pi_k}(s) \leq V_{\pi_{k+1}}(s), \forall s \in \mathcal{S}$ where k denotes the policy update episode index. The value function is bounded by $V_{\max}$. Therefore, the sequence of value function $(V_{\pi_k})_{k \in \mathbb{N}}$ converges. This is limited by V .

Step 2 (Characteristic of limit points): When using an iterative algorithm in the learning process to find an NE, the algorithm generates a sequence of strategy profiles that gradually converge to a limit point. This limit point represents the strategy profile reached at infinite iterations and is crucial for identifying the NE during learning [40]. Since the joint policy space Π is bounded, allowing the application of the Bolzano-Weierstrass theorem [41] to ensure the existence of a convergent subsequence $(\pi_k)_{k \in \mathbb{N}}$ and thus a limit point. Let $\bar{\pi}$ to be such limit point. We introduce the following notation: for a joint policy π and a permutation $b_{1:M}$, the $HU(\pi, b_{1:M})$ be a joint policy obtained by a policy update from π along with the permutation $b_{1:M}$.

Claim: For any permutation $z_{1:M} \in \text{Sym}(M)$, the following equation holds $\bar{\pi} = HU(\bar{\pi}, z_{1:M})$.

The proof of this claim can be found in [39].

Step 3 (Dropping the drift function): As introduced before, the proof and policy optimization are based on the ML framework. The drift function is designed to stabilize the training progress. This step drops the drift function to prove that the pure policy update can still converge to the NE. With the claim in Step 2, for any permutation $b_{1:M}$, considering the first component of HU, $\bar{\pi}^{b_1}$, the following equation can be obtained:

$$\bar{\pi}^{b_1}(\cdot^{b_m} | s) = \arg \max_{\pi^{b_1} \in \mathcal{P}(\mathcal{A}^{b_1})} \mathbb{E}_{a^{b_1} \sim \pi^{b_1}} [Q_{\bar{\pi}}^{b_1}(s, a^{b_1})]. \quad (36)$$

Compared with (35), the drift function in (36) is dropped. Since the drift function is non-negative, and at $\pi^{b_1} = \bar{\pi}^{b_1}$, its value, and of all its Gâteaux derivatives are zero.

Step 4 (Converging to Nash equilibrium): The policy $\bar{\pi}$ satisfies

$$\begin{aligned} \bar{\pi}^b(\cdot^b | s) &= \arg \max_{\pi^b(\cdot^b | s) \in \mathcal{P}(\mathcal{A}^b)} \mathbb{E}_{a^b \sim \pi^b} [Q_{\bar{\pi}}^b(s, a^b)] \\ &= \arg \max_{\pi^b(\cdot^b | s) \in \mathcal{P}(\mathcal{A}^b)} \mathbb{E}_{a^b \sim \pi^b, a^{-b} \sim \bar{\pi}^{-b}} [Q_{\bar{\pi}}(s, a)]. \end{aligned} \quad (37)$$

Equation (37) is based on the finding with **Definition 2** and **Lemma 1** that the update policy $\bar{\pi}^i$ for agent i are optimized. When the remaining agent actions are made, the best response is provided under the decisions of the rest of the agents. This procedure can be solved by the iteration method or sequential update, as we mentioned in Fig. 3. Hence, if the joint policy from the rest of the agents $\bar{\pi}^{-b}$ is fixed, we can conclude that the policy $\bar{\pi}^b$ in equation (37) satisfies the condition for the optimal policy. Therefore, the following equation can be obtained:

$$\bar{\pi}^b = \arg \max_{\pi^b \in \Pi^b} J(\pi^b, \bar{\pi}^{-b}). \quad (38)$$

where $\bar{\pi}$ is the NE. Meanwhile, it also implies that the value function corresponds to a Nash value function V^{NE} , the return is corresponding to a Nash return J^{NE} . The four-step proof matches **Definition 1**, and this training framework with theoretical properties can guide agents' policy updates to approach NE. More details for discussion and analysis regarding the proof can be found in the supplementary document [38].

F. Casting on Trust Region Policy Optimization

With the NE discovery training framework provided in Section IV-E, we aim to use an instance for this framework and cast this framework into a practical algorithm. Trust region policy optimization (TRPO) [42] naturally inherits the ML property. For instance, the KL-divergence is used in TRPO to stabilize the training between old and new policies. It is similar to the property of neighboring operators in the ML framework. On the other hand, for the multi-agent bidding problem, the on-policy algorithm such as TRPO can learn from the highly dynamic trading environment from a real-time bidding trajectory. The ML framework is designed as follows:

Drift function: 0

Neighboring operator: $\mathcal{U}_{\pi_k}^b(\pi_k^b) = \{\pi^b \in \Pi^b \mid \mathbb{E}_{s \sim \rho_{\pi_k}} [D_{\text{KL}}(\pi_k^{b_m}(\cdot | s), \pi^{b_m}(\cdot | s))] \leq \delta\}$.

Policy updates target: $\pi_{k+1}^{b_m} = \arg \max_{\pi^{b_m}} \mathbb{E}_{a^{b_{1:m-1}}, \pi_{k+1}^{b_{1:m-1}}, a^{b_m} \sim \pi^{b_m}} [A_{\pi_k}^{b_m}(s, a^{b_{1:m-1}}, a^{b_m})]$.

where the neighboring operator is modeled as KL-divergence constraints. The details can be found in TRPO [42]. The policy is updated sequentially as we introduced in the ML framework. Then, we summarize how the actor-network and critic-network are updated.

Update in actor-network: The actor-network is parameterized by θ . The update rule can be written as in the k th update episode:

$$\theta_{k+1}^{b_m} = \arg \max_{\theta^{b_m}} \mathbb{E}_{\underbrace{a^{b_{1:m-1}} \sim \pi(\theta)_{k+1}^{b_{1:m-1}}}_{\text{Updated}}, \underbrace{a^{b_m} \sim \pi(\theta)_k^{b_m}}_{\text{Ready to update}}}$$

Obj:

$$A_{\pi(\theta)_k^{b_m}}^{b_m}(s, a^{b_{1:m-1}}, a^{b_m}) \quad (39)$$

$$\text{St. } \mathbb{E}_{s \sim \rho_{\pi(\theta)_k}} [D_{\text{KL}}(\pi(\theta)_k^{b_m}(\cdot | s), \pi(\theta)^{b_m}(\cdot | s))] \leq \delta \quad (40)$$

where δ denotes the threshold for stable policy updates.

Each prosumer performs sequential updates during their respective policy optimization processes. This ensures that when updating their strategies, they consider the strategies of others. Meanwhile, we can observe that $a^{b_{1:m-1}} \sim \pi(\theta)_{k+1}^{b_{1:m-1}}$ is labeled as the policy updated. In this term, the joint action a and policy π is bold, it indicates the policy network of more than one agent has been updated. On the other hand, the term $a^{b_m} \sim \pi(\theta)_k^{b_m}$ denotes the policy ready to be updated. It can be inferred that the policy update for parameter network θ^{b_m} depends on the update from the rest of the agents $b_{1:m-1}$. It should also be highlighted that with the **Lemma 1** for the advantage decomposition, the joint optimization target can be easily decomposed to the agent b at permutation index m and policy update iteration k . The sequential update procedure is implemented as in Fig. 3. In the real implementation stage, the sequential update for agent permutation can be done by torch.randperm() function from pytorch library. Although we provide the NE discoverable policy update rule in (39), it is still hard to compute directly; linear approximation is needed for the objective function, and quadratic approximation is required for KL constraints in (40). Then, the policy optimization model in (39)–(40) can be solved by the following closed-form update rule from iteration k to $k+1$ as in equation (41):

$$\theta_{k+1}^{b_m} = \theta_k^{b_m} + \alpha \sqrt{\frac{2\delta}{\mathbf{g}_k^{b_m} (\mathbf{H}_k^{b_m})^{-1} \mathbf{g}_k^{b_m}}} (\mathbf{H}_k^{b_m})^{-1} \mathbf{g}_k^{b_m}. \quad (41)$$

where $\mathbf{H}_k^{b_m}$ is the hessian matrix of KL-divergence return, $\mathbf{g}_k^{b_m}$ indicates the gradient of (39), and α is the learning coefficient that can be tuned by the backtracking line search. Furthermore, estimating the objective in (39)–(40) is challenging since we introduce the sequential update rules that previous agents'

TABLE I
PROSUMER TYPES AND LOCATIONS

Prosumer No.	Prosumer type and node index(phase)	Prosumer No.	Prosumer type and node index(phase)
1	Load dominated 8(a)	6	PV dominated 80(b)
2	Load dominated 52(a)	7	PV dominated 99(c)
3	Load dominated 76(b)	8	WT dominated 89(c)
4	Load dominated 62(a)	9	Balanced 43(c)
5	PV dominated 50(b)	10	Balanced 29(c)

decisions need to be considered. The importance sampling techniques can be applied for gradient computation and then optimized by conjugate gradient algorithm. More details for the TRPO update can be found in [42].

Update in critic network: The critic network parameterized by ϕ is designed for prosumers' policy evaluation. Then, one centralized network is introduced. In this algorithm, the critic network update rule is designed as in (42) with the loss function by collecting trajectory data samples of size BT.

$$\phi_{k+1} = \underset{\phi}{\operatorname{argmin}} \frac{1}{BT} \sum_{B=1}^{BT} \sum_{t=0}^T (V_{\phi}(s_t) - R_t)^2 \quad (42)$$

V. CASE STUDY

A. Test System, Comparable Algorithms and Methods

1) *Test System:* The proposed method is tested on a modified IEEE 123-bus unbalanced distribution system, with the integrated prosumers' type, location and DER portfolio details summarized in Table I. The bus voltage upper and lower limits are set as 0.95 p.u. and 1.05 p.u., respectively. The initial carbon emission cap $c_{b,t}^{\text{cap}}$ is set as 0.05 tonsCO₂ for each prosumer. The basic carbon permit cap from DSO $\bar{C}_{j,t}^{\phi}$ for each phase is as 0.10 tonsCO₂. The carbon excessive price $\hat{C}_{j,t}^{\phi}$ is obtained from [43]. We use 90 \$/tonsCO₂ here. The carbon intensity for MT is set as 0.8 tonsCO₂/MWh, and for RES, it varies between 0.1 and 0.2 tonsCO₂/MWh, depending on the location and device capacity. The DSO-owned DER location and capacity can be found in [31]. The price $\lambda_{0,t}$ at the slack bus from the upper level market is sourced from [19]. All prosumers are equipped with MT and demand response, and the load. The MT output range is set as [0,0.02] MWh, and the maximum demand response ratio $\varepsilon_b^{\text{dr}}$ is set as 10% of the load for each prosumer.

The RES generation profile with prediction error methods are sourced from [44]. The cost function coefficient for MT and DR and technique parameters for prosumers' can be found in [3].

2) *Baseline Algorithms:* The DRL environment is implemented using Gym, and the neural network training is conducted with PyTorch. The CAOPF is solved by the Gurobi solver. The proposed NE discovery algorithm is named **NE-TRPO**. Its performance is benchmarked against several baseline algorithms: The MADDPG algorithm in [16], the state-of-the-art cooperative algorithm MAPPO [15], the TRPO algorithm without NE training framework, namely Vanilla-TRPO naïvely extended to the MADRL framework.

The comparison of P2P trading methods is listed below: **M1:** The carbon emission permits trading is not considered

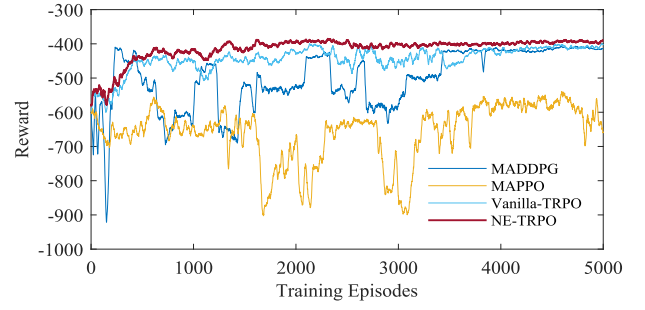


Fig. 4. Rescaled rewards of DRL algorithms.

TABLE II
PERFORMANCE COMPARISON OF ALGORITHMS

Algorithm	Reward and variance	Converge Episodes
MADDPG	-416±4.6	4107
MAPPO	-585±27.2	-
Vanilla-TRPO	-408±4.1	3410
NE-TRPO	-395±3.2	2610

in the CDA market. **M2:** The carbon emission regulation from DSO is not considered. **M3:** The single-phase network approximation is applied rather than an unbalanced network. **M4 (Proposed):** JECT in CDA is considered, DSO is responsible for carbon emission regulation, and unbalanced network modeling is considered.

B. Superiority of NE Discovery TRPO

The proposed NE-TRPO is compared with the existing method in Fig. 4. The reward indicates the sum of cost for all prosumers in this game.

From Fig. 4, we can observe that the proposal converges in 2600 episodes, which is 36.39% shorter than the MADDPG algorithm. The MAPPO algorithm is even hard to converge with the 5000 episodes in this complex coordination task. The results indicate fast convergence capabilities of the proposal with theoretically guided training. Meanwhile, the proposed NE-TRPO reduces the reward by 32.48% than MADDPG and 32.5% than MAPPO. Compared with the Vanilla-TRPO, the NE-TRPO enhances the reward by 3.29% and convergence time by 23.46%. Table II further compares the stability of different methods through the evaluation of reward variance. The proposed training framework is shown to demonstrate the best training stability. The results confirm the effectiveness of the sequential update framework.

Besides the empirical training performance, the proposed NE-TRPO with theoretical properties further underpins the empirical performance. Each prosumer is satisfied with this coordination and provides the best response to the rest of the agents in the game, hindering the superior stable coordination performance. The following test will be implemented with trained NE-TRPO agents.

C. Bidding Analysis in Joint Electricity and Carbon Trading

Then, the bidding results for a one-day operation are plotted in Fig. 5, and the P2P JECT and DSO price are plotted. Since the price provided by DSO is spatial, temporal coupling

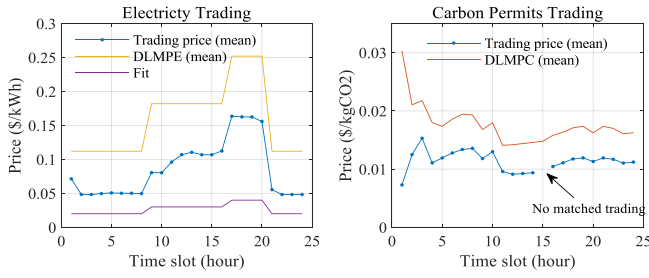


Fig. 5. Bidding results in JECT market.

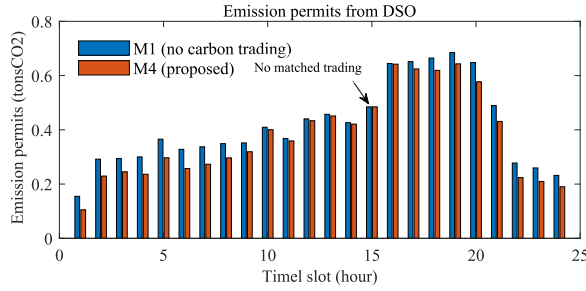


Fig. 6. Carbon permits from DSO.

TABLE III
PERFORMANCE COMPARISON OF METHODS

Methods	M1	M2	M4
Cost (\$)	1501	1561	1553
Carbon Emission (tonsCO ₂)	21.30	20.79	20.79

according to the location of prosumers in the network, we used the mean price here for better illustration and aligning with the bidding range in equation (22) and (23). From the left side of Fig. 5, we can observe that the cleared price falls within the range between the DLMPE and the Fit. This price range creates a margin for P2P agents to profit from their trades. Meanwhile, the NE-TRPO allows for various matched trades based on bidding decisions. For the right-side carbon trading, the DLMPC decreases from 10.00 to 16.00, attributed to the high PV generation output during this period. Low nodal carbon intensity results in a lower DLMPC, which aligns with modelling in (18).

Additionally, at 15:00, no carbon trading quantity was matched, indicating that all prosumer agents were required to purchase emission permits from the DSO. This suggests that all prosumers exceeded their local emission caps, leaving no surplus permits to trade within the P2P platform. However, even with unmatched quantities in a single time slot, prosumers can still purchase fewer carbon permits from the DSO throughout the day, leading to further cost reductions as in Fig. 6. As summarized in Table III, this approach saves \$8 in operational costs over a single day, without increasing carbon emissions. It is also important to note that, compared to the electricity market, the DLMPC in the carbon market is more stochastic and dynamic. The data-driven NE-TRPO effectively captures this market volatility, enabling intelligent bidding decisions. Finally, it can be concluded that the successful

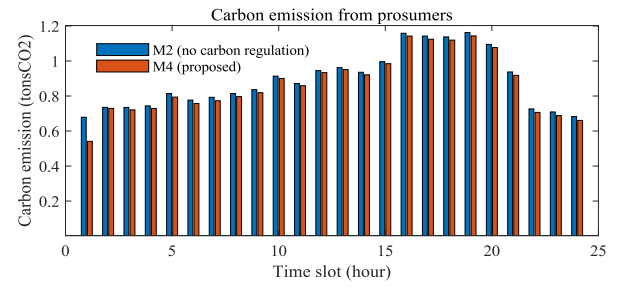


Fig. 7. Carbon emission comparison between M2 and M4.

integration of the JECT CDA market can further reduce operational costs without increasing carbon emissions.

D. Results of Carbon-Aware P2P Energy Trading

We further discuss how this trading framework can mitigate carbon emissions for environmentally friendly trading in Fig. 7.

In this section, the **M2** is used for analyzing that the carbon regulation is not considered. Since DSO is not allowed to restrict individual users to their carbon emissions, system-level tracing and accounting and market clearing are efficient methods to provide incentives for emission regulation. It means that they are encouraged to use the high carbon intensity device less to save the cost.

As shown in Fig. 7, with the cleared price for incentive, the M4 produces less carbon emission for each time slot. It can also be observed in Table III that 0.51 tonsCO₂ is less produced compared to **M2**, verifying the effectiveness of emission reduction. These results indicate that the proposed method can reduce costs and carbon emissions simultaneously.

E. Sensitivity Analysis for Local Carbon Cap

Although there are currently no baseline and grid code for emission constraints, we apply the sensitivity analysis for the local carbon emissions cap against the emission credit purchased from the DSO. The step size for the local cap is selected as 0.01 tonsCO₂ based on the cap of 0.05 tonsCO₂. The test is implemented based on the trained NE-TRPO agents. The horizontal axis represents the carbon cap for each local prosumer per hour, while the vertical axis represents the total carbon emission permits purchased by all prosumers from DSO over one day operation. From Fig. 8, we can observe that when the carbon cap is less than 0.05 tonsCO₂, prosumers tend to purchase more carbon permits from the DSO, as most local prosumers no longer have space for further transactions (i.e., smaller carbon caps lead to less surplus emission permits). Conversely, when the local carbon cap exceeds 0.05 tonsCO₂, prosumers reduce their purchases of carbon permits from the DSO. Combined with the findings in Fig. 6 and Table III, more emission credits purchased from DSO results in less emissions and higher operation costs. Therefore, we claim in the manuscript that the local cap can be considered as the system input parameter. At the real time implementation stage, the local P2P market operator can determine this value to balance cost reduction and emission regulation.

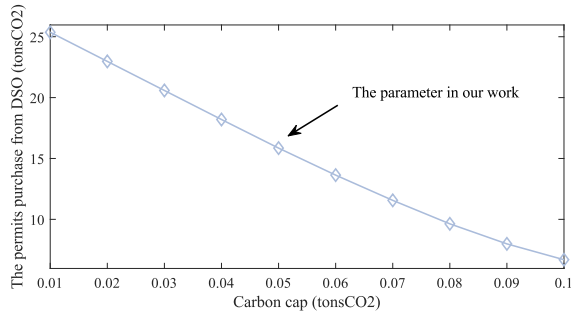


Fig. 8. The sensitivity analysis for local carbon cap.

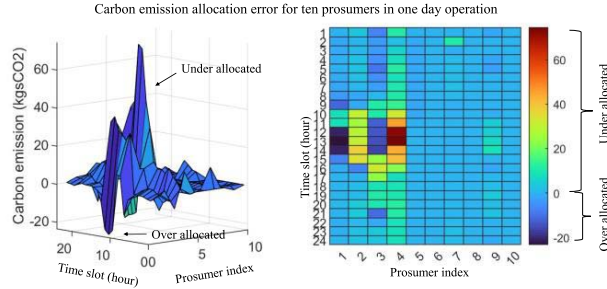


Fig. 9. Emission accounting errors of single-phase approximation.

F. Significance of Carbon Allocation in Unbalanced Network

We also compare the results when using a single-phase approximation, as similarly applied in [3]. There are two conditions for this approximation. In **Condition I**, we approximate the unbalanced network when the MT and load type prosumer are located at the same node on different phases. Due to the proximity of the MT, carbon intensity is more concentrated, leading to an ‘over-allocation’ of emissions. In **Condition II**, the RES and generation type prosumer connect at the same node in different phases. Applying the single-phase approximation in this case dilutes the carbon intensity at the node, resulting in ‘under-allocation.’ These two conditions present fairness issues for trading operations. In the case of ‘over-allocation,’ more emissions are attributed to the prosumer, negatively impacting their economic benefits. Conversely, ‘under-allocation’ leads to fewer emissions being attributed to the prosumer, which is unfair to the DSO when clearing service fees.

This allocation error was plotted in Fig. 9. On the left side, we can observe that over and under-allocation exists for each prosumer in a one-day operation. If the error exceeds zero, the over-allocation exists, and vice versa. The heatmap on the right side clearly plots the prosumers’ level emission allocation error caused by single-phase approximation. Prosumer 1 at 12:00 to 14:00 suffers from over-allocation, and Prosumer 4 at time 12:00 to 14:00 exists serve under-allocated. The allocation error happens heavily at noon since the low carbon intensity PV output may dilute/concentrate the emissions accounting. This leads to unfair operations for both DSO and prosumers. Table IV further summarizes cost difference due to the accounting error from 12:00 to 14:00 for Prosumers 1-4. The results imply that ‘over-allocated’ increases the cost

TABLE IV
PERFORMANCE COMPARISON OF METHODS – PROSUMER SIDE

Meth- ods/agent (cost \$)	Prosumer1	Prosumer2	Prosumer3	Prosumer4
M4	44	43	41	143
M3	47	40	40	136

(Prosumer 1) and influences the trading interest of prosumer 1. On the other hand, ‘under-allocated’ may decrease the cost for prosumers, and DSO will lose the profits. Unbalanced network modeling leads to a more accurate emission allocation for carbon-aware P2P trading.

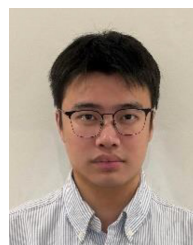
VI. CONCLUSION AND FUTURE WORK

This paper proposes a carbon-aware trading mechanism for prosumers in an unbalanced network. The NE-TRPO method enhances the theoretical foundation with Nash Equilibrium discovery, providing both reliability and interpretability for trading strategies. Additionally, its performance surpasses that of various MADRL algorithms. The JECT CDA market offers trading flexibility with limited market information and straightforward peer matching, leading to cost reductions and fewer emission permits purchased from the DSO. By clearing carbon-aware prices within the unbalanced network, the DSO ensures accurate emission allocation and promotes prosumers’ environmentally friendly trading decisions that help mitigate carbon emissions. Future work will focus on extending this NE discovery method to additional algorithms for various power system operations and comply the voltage unbalance factor constraint into the operational model.

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